

Comparative Analysis of Meta-Heuristic Algorithms for Multi-Robot Search and Rescue

Nada Ehab*, Farah Khaled[†], Martin Morcos*, Ibrahim Abdelaty*,
Abdelrahman Y. Altaher*[‡], Omar M. Shehata*[‡]

* Mechatronics Department, Faculty of Engineering and Materials Science

[†] CSEN Department, Faculty of Media Engineering and Technology

[‡] Multi-Robot Systems (MRS) Research Lab

German University in Cairo, Egypt

(nadaehab2003, farahkhaled03, martin.morcos, labdelatyeng@gmail.com,

omar.shehata@ieee.org, Abdulrahmanyasserli@gmail.com

Abstract—We present a comparative analysis of four meta-heuristic optimization algorithms applied to the Multi-Robot Search and Rescue (MRSAR) problem: Simulated Annealing (SA), Genetic Algorithm (GA), Ant Colony Optimization (ACO), and Harris Hawks Optimization (HHO). The objective of the algorithms was to minimize the cost function of the rescue operation, made up of the robot's travel distance and the adherence to the rescue time window. Experimental results indicate that the Genetic Algorithm (GA) achieved the best convergence and final solution quality (Cost approximately 103), followed closely by Harris Hawks Optimization (HHO).

Index Terms—Multi-robot, Optimization Algorithms, Search and Rescue, Path Planning

I. INTRODUCTION

Multi-Robot Path Planning (MRPP) is a fundamental and computationally challenging problem in robotics that requires simultaneous consideration of route feasibility, collision avoidance, temporal constraints, and high-level coordination. MRPP instances are considered NP-hard which means as the number of robots, obstacles, and time-window constraints grow, exact methods become difficult to manage. In addition, practical solvers must trade optimality for robustness and run-time performance which affects the viability of the algorithms used. In real-world applications — from warehouse automation to search-and-rescue and environmental monitoring — effective MRPP must be ensured. Effective MRPP must balance path length, computational efforts, and solution reliability while enduring the limitations faced such as limited and complex computational resources in clustered environments.

Meta-heuristic optimization algorithms provide a flexible, successful number of approaches for MRPP because they can explore large, multi-modal search spaces, escape local minima, and adapt to multiple objective criteria. This report utilizes four representative meta-heuristics techniques to solve a search and rescue problem: Simulated Annealing (SA), Genetic Algorithm (GA), Ant Colony Optimization (ACO), and Harris Hawks Optimization (HHO), all in a controlled 25×25 grid environment with static obstacles that are randomly generated in each run of the program, evaluating each method against a composite cost function that combines travel distance and a time-window for each victim.

The goal of this paper is divided into two stages: first, a rigorous, side-by-side empirical comparison of different meta-heuristic techniques when applied to MRPP while maintaining identical simulation conditions is provided; and second, algorithm selection between population-based, trajectory-based, and swarm-based approaches and tuning the parameters of the selected algorithm. The structure of the paper follows a sequence of explaining the parameters definition, simulation environment, definition of each algorithm, comparative analysis and interpretation of results, and a comparison of the performance of each algorithm.

II. LITERATURE REVIEW

Research in recent years view Search and Rescue (SAR) problems using Multi-Robot Systems (MRSs) as a set of optimization problems involving task allocation, coverage, and path planning under a set of operational constraints. An aspect most papers focus on is the decomposition of the problem, as well as solution design, where large, complex SAR missions are broken into manageable sub-problems and solved using a set of optimization techniques, heuristic search, metaheuristics, and decentralized control. Objectives typically balance competing goals such as maximizing coverage and detection probability, minimizing time and energy expenditure, ensuring workload balance among robots, and maintaining robustness under limited communication and sensing capabilities.

Cai et al. [1] formulate maritime SAR with Autonomous Underwater Vehicles (AUVs) as a Multi-robot Coverage Path Planning (MCP) problem decomposed into area partitioning and single-AUV path planning. The area partitioning stage optimizes the ordering of AUVs and split-line angles to minimize segmentation imbalance and balance workload while ensuring a full, non-overlapping coverage of the search area. The path planning stage then generates sonar-efficient trajectories that prioritize high-probability target regions while minimizing turns. The authors solve these sub-problems using a customized backtracking algorithm inspired by the N-Queens problem and a modified A* search (SAR-A) that incorporates distance, target discovery probability, and turning cost.

Osooli et al. [2] address Multi-Robot Task Assignment (MRTA) for heterogeneous SAR teams through a multi-stage optimization framework. Binary decision variables represent robot-to-cluster and robot-to-victim assignments, with objectives that maximize fulfillment of victim requirements and minimize expected task completion time while accounting for success and failure probabilities. The imposed constraints ensure exclusive assignment of robots to clusters and victims. Their proposed Multi-Stage Multi-Robot Task Assignment (MSMRTA) framework combines linear assignment methods with market-based auctions and integrates A* path planning, enabling scalable and reliability-aware coordination in complex SAR scenarios.

Kashyap et al. [3] focus on multi-robot exploration in unknown urban SAR environments, formulating the problem around the collective motion of a robot swarm under severe constraints, including lack of global positioning, limited communication, and unknown obstacles. The objective is to maximize explored area within a fixed time horizon. To achieve this, the authors propose a hybrid Levy Flight-Particle Swarm Optimization (LF-PSO) algorithm, where Levy Flight helps increase the exploratory behavior of the robots, while the repulsion-based social component reduces overlap between robots paths, promoting efficient dispersion and coverage without centralized coordination.

Baxter et al. [4] investigate a decentralized, reactive approach to multi-robot search based on artificial potential fields. In this formulation, each robot's motion is guided only by local sensor data and shared potential field information from nearby robots, with the objective of minimizing the time to discover all targets. The key constraint introduced in this paper is the absence of deliberative planning, as robots rely only on real-time reactions. The authors propose both pessimistic and optimistic potential field sharing strategies, which differ in how conservatively robots respond to shared obstacle information, emphasizing the trade-offs between safety and search efficiency in the robots.

Finally, Drew [5] provides a system-level review of multi-agent systems for SAR, emphasizing the gap between academic research and reliable real-world deployment. Rather than defining a single optimization problem, the paper identifies overarching constraints such as scalability, unreliable communication, difficult human-robot interaction, and cost-capability trade-offs. Drew focuses on research directions, including distributed SLAM, optimization- and market-based task allocation, ad-hoc mesh networking, and sliding autonomy models.

These mentioned works underscore that effective SAR with MRSs requires not only algorithmic optimization but also robust integration across perception, communication, planning, and human supervision to achieve field-ready systems.

This work aims to provide a comparative analysis between some of the state-of-the-art optimization algorithms, and how they could be utilized in the SAR operations to provide maximum rescue chance for each victim while maximizing the number of rescued victims.

III. METHODOLOGY

In our research, we focus on comparing different optimization techniques within the Multi-Robot Search and Rescue (MRSAR). Multiple test scenarios were simulated, however, any scenario where a collision occurred was considered an infeasible solution.

A. Parameters Definition

Our objective functions are comprised of two main points, minimizing the distance traveled by each robot, as well as maximizing the number of rescued victims compared to the number of missed victims. We defined three decision variables, the path to be taken by each robot, the order of points within the path to be taken by each robot, as well as the time window within which the victim has to be rescued by the robot before the task expires. Four constraints were applied to our problem, the number of available robots to perform the rescue, the obstacles within the environment, the battery capacity of each robot in the fleet, as well as the time window the robot has to perform the rescue before the mission is missed. The battery capacity of the robot is modeled to affect the velocity of the robot. Whenever the battery charge decreases, velocity of the robot decreases linearly with it. In order to imitate this scenario in the simulation, the battery capacity was decreased by a certain factor every time step.

All techniques used had the same problem formulation, as shown in the following equations:

$$Distance = \sum_{i=0}^G |n_{i+1} - n_i| = D_1 + D_2 + \dots + D_G$$

where n_i is the node the robot is currently located, and n_{i+1} is the node the robot will be heading towards.

$$Rescue\ Chance = \sum_{i=0}^R (TW - \frac{\sum_{i=0}^G |n_{i+1} - n_i|}{V_R}) = \sum_{i=0}^R (TW_R - T_R)$$

where the R is the number of available robots, V_R is the robot's speed which is determined by the robot's battery capacity and TW is the time window for each victim. Once the rescue chance value is computed using the above equation, the parameters called missedVictims and savedVictims are updated. Our target function is to increase the savedVictims with respect to the missedVictims while minimizing the distance taken.

The outcome is calculated as shown in the following equation:

$$Outcome(r) = \begin{cases} \text{savedVictims} + 1, & \text{if } r \geq 0 \\ \text{missedVictims} + 1, & \text{if } r < 0 \end{cases}$$

; where r is the rescue chance.

B. Environment Setup

The simulation parameters used for all algorithms are detailed in Table I.

TABLE I
SIMULATION ENVIRONMENT PARAMETERS

Parameter	Value
Grid Size	25×25
Number of Robots	5
Obstacle Density	$\approx 13\%$
Robot Velocities	$0.8 - 1.5$ units/step
Weights (w_{dist}, w_{time})	(1.0, 0.5)
Max Iterations	600 (500 for ACO)

C. Algorithms

The algorithms we covered within our research were Simulated Annealing (SA), Genetic Algorithm Optimization (GA), Ant Colony Optimization (ACO) [6], and Harris Hawks Optimization (HHO) [7] techniques. SA is a stochastic optimization heuristic that iteratively refines candidate solutions while probabilistically accepting worse states to avoid a local minima. A temperature parameter controls the transition from broad exploration to focused exploitation. SA is used to improve multi-agent task or path assignments under uncertainty by guiding the system toward a near-optimal global configuration. GA is an evolutionary search method that operates on a population of candidate solutions encoded as chromosomes. Through selection, crossover, and mutation, the algorithm evolves to higher-quality solutions. Within the multi-agent Search and Rescue context, the GA enables efficient exploration of large combinatorial spaces and supports robust optimization of task allocation and coordination strategies. ACO is a swarm-based metaheuristic inspired by pheromone-mediated path finding in ant colonies. Artificial ants construct solutions probabilistically, guided by pheromone intensities that encode solution quality over iterations. For the multi-agent system, ACO facilitates decentralized route planning and promotes the emergence of efficient search patterns in complex environments. Harris Hawks Optimization is a nature-inspired algorithm that models the cooperative, surprise-based hunting strategies of Harris hawks. The method adaptively alternates between exploration and exploitation based on the prey's modeled escape energy. In this study, HHO is applied to optimize multi-agent deployment strategies, enabling effective convergence in nonlinear and highly dynamic search spaces.

IV. RESULTS AND COMPARISON

To account for the stochastic nature of the tested metaheuristics and the random generation of static obstacles in the environment, each algorithm was evaluated over 18 independent runs. The convergence plots shown in this section correspond to representative runs and are included to illustrate the search behavior of each method. However, all performance conclusions and rankings are based on the statistical analysis over the full 18-run dataset rather than on a single run.

A. Simulated Annealing

Simulated Annealing (SA) showed a gradual reduction in cost, but its performance exhibited noticeable variability across runs. In the representative run shown in Fig. 1, the algorithm escaped poor initial solutions and then converged to a plateau.

Over the 18 runs, SA achieved a mean final cost of 154.25 ± 23.64 and a mean execution time of 22.92 ± 23.16 s.

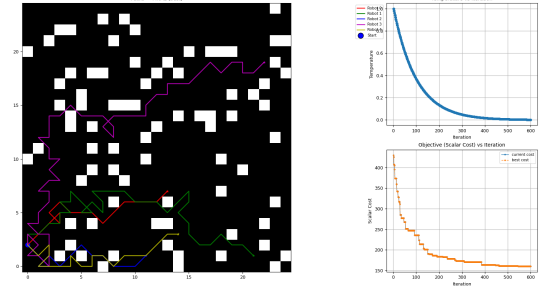


Fig. 1. Representative run for Simulated Annealing. The left side shows the generated robot paths, while the right side shows the cost and temperature evolution over the iterations.

B. Genetic Algorithm

The Genetic Algorithm (GA) achieved the best overall solution quality among all tested methods. As illustrated by the representative run in Fig. 2, GA maintained steady improvement throughout the search due to the diversity preserved in the population. Across the 18 runs, GA obtained the lowest mean final cost, equal to 107.78 ± 11.23 , with a mean execution time of 117.37 ± 129.13 s.

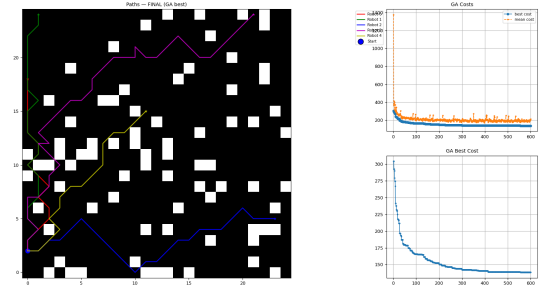


Fig. 2. Representative run for the Genetic Algorithm. The left side shows the generated robot paths, while the right side shows the evolution of the mean and best cost over the iterations.

C. Ant Colony Optimization

Ant Colony Optimization (ACO) exhibited fast initial improvement, but it frequently stagnated at higher-cost solutions and showed the highest computational burden. The representative run in Fig. 3 reflects this behavior, where the cost decreases early and then stabilizes. Over all 18 runs, ACO achieved a mean final cost of 146.39 ± 25.73 and the highest mean execution time of 790.21 ± 1108.73 s.

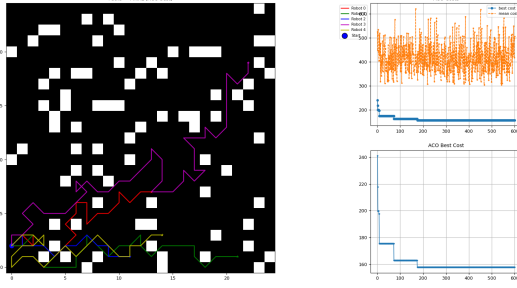


Fig. 3. Representative run for Ant Colony Optimization. The left side shows the generated robot paths, while the right side shows the evolution of the mean and best cost over the iterations.

D. Harris Hawks Optimization

Harris Hawks Optimization (HHO) produced the second-best overall performance. Its representative convergence profile in Fig. 4 shows step-like improvements, which indicates the algorithm's ability to escape local optima and continue refining the solution. Across the 18 runs, HHO obtained a mean final cost of 118.11 ± 18.67 and a mean execution time of 238.76 ± 374.91 s.

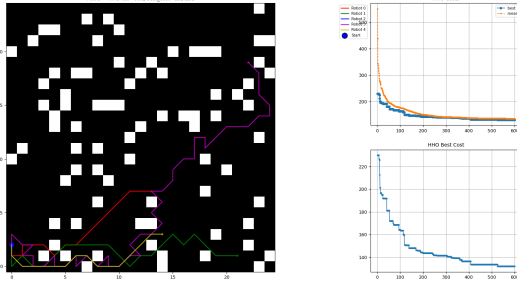


Fig. 4. Representative run for Harris Hawks Optimization. The left side shows the generated robot paths, while the right side shows the evolution of the mean and best cost over the iterations.

E. Statistical Analysis Over 18 Independent Runs

Table II summarizes the final cost and execution time of all algorithms over the 18 runs. Lower cost indicates better path-planning and rescue-task performance.

TABLE II
STATISTICAL SUMMARY OVER 18 INDEPENDENT RUNS

Algorithm	Final Cost (Mean $\pm$ SD)	Execution Time [s] (Mean $\pm$ SD)
SA	154.25 $\pm$ 23.64	22.92 $\pm$ 23.16
GA	107.78 $\pm$ 11.23	117.37 $\pm$ 129.13
ACO	146.39 $\pm$ 25.73	790.21 $\pm$ 1108.73
HHO	118.11 $\pm$ 18.67	238.76 $\pm$ 374.91

To visualize the distribution of the final cost values, a box-and-whisker plot is provided in Fig. 5. The figure confirms that GA not only achieved the lowest average cost, but also maintained the tightest spread among the best-performing methods, indicating more reliable convergence. HHO ranked

second, while ACO and SA showed higher costs and broader dispersion.

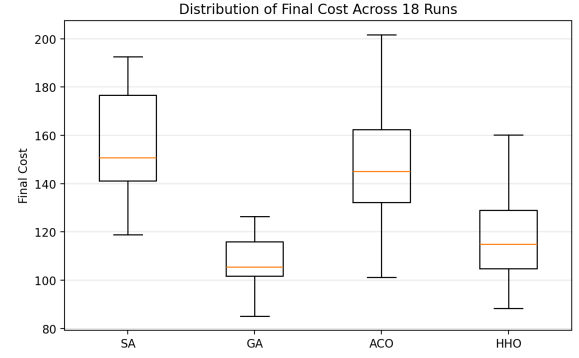


Fig. 5. Box-and-whisker plot of the final cost obtained by each algorithm over 18 independent runs.

An overall Friedman test was first conducted on the final-cost results of the four algorithms and confirmed that the differences among the methods were statistically significant ($\chi^2(3) = 48.87$, $p = 1.39 \times 10^{-10}$). Since the runs were treated as matched scenarios across all algorithms, pairwise comparisons were then performed using the Wilcoxon signed-rank test with Holm correction. The results are shown in Table III.

TABLE III
PAIRWISE WILCOXON SIGNED-RANK TEST RESULTS FOR FINAL COST (HOLM-CORRECTED)

Comparison	p -value	Result
GA vs. HHO	0.0012	Significant
GA vs. ACO	0.0012	Significant
GA vs. SA	0.0012	Significant
HHO vs. ACO	0.0012	Significant
HHO vs. SA	0.0012	Significant
ACO vs. SA	0.0248	Significant

F. Algorithm Results Comparison

Figure 6 shows a representative convergence comparison for one run. It can be observed that ACO improves rapidly at the beginning, GA maintains the strongest long-term refinement, and HHO demonstrates occasional jumps that help it escape local minima. Nevertheless, this figure is only illustrative; the final ranking is based on the 18-run statistical analysis presented above.

Based on the mean final cost over the 18 independent runs, the algorithms are ranked as follows:

- 1) **Genetic Algorithm (GA):** 107.78 ± 11.23
- 2) **Harris Hawks Optimization (HHO):** 118.11 ± 18.67
- 3) **Ant Colony Optimization (ACO):** 146.39 ± 25.73
- 4) **Simulated Annealing (SA):** 154.25 ± 23.64

These results show that GA provided the best solution quality and significantly outperformed the other three methods in terms of final cost. HHO achieved the second-best performance and also significantly outperformed both ACO

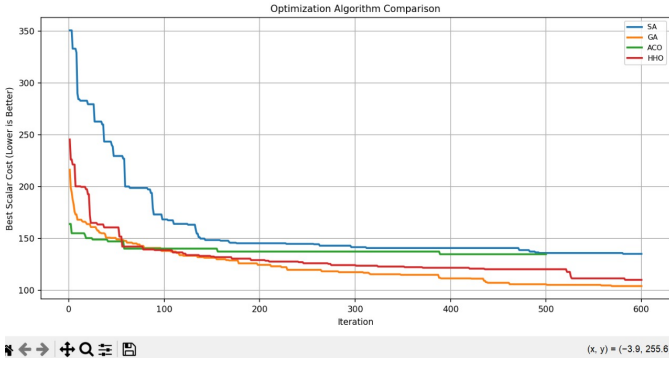


Fig. 6. Representative convergence comparison of the four algorithms for one run.

and SA. Although SA had the lowest average execution time, it produced the highest final cost among the tested approaches. In contrast, ACO was computationally the most expensive while still yielding weaker solution quality than GA and HHO.

V. CONCLUSION

This paper presented a comparative study of four meta-heuristic optimization algorithms—Simulated Annealing (SA), Genetic Algorithm (GA), Ant Colony Optimization (ACO), and Harris Hawks Optimization (HHO)—applied to a Multi-Robot Search and Rescue (MRSAR) problem in a 25×25 grid environment with static obstacles and time-window constraints. All algorithms were tested under the same conditions using a unified cost function that measured both total travel distance and how well each robot met its rescue time window.

The results showed that the Genetic Algorithm (GA) provided the best overall performance, achieving the lowest final cost (about 103.4) and maintaining steady improvement throughout the search. Harris Hawks Optimization (HHO) produced the second-best result (about 109.9) and showed clear jumps in performance, demonstrating its ability to escape poor solutions and improve further. Ant Colony Optimization (ACO) improved quickly at the beginning but slowed down early and could not reach a better solution later, finishing with a higher cost (about 134.8). Simulated Annealing (SA) also improved over time but eventually stopped making progress, ending with a final cost similar to ACO (about 135.2).

Overall, the findings indicate that population-based algorithms, especially GA, are more effective for solving multi-robot path planning problems with time constraints because they explore the search space more thoroughly and find more reliable solutions. Future work may include testing hybrid approaches (such as combining GA with HHO or adding local search to ACO), applying the algorithms in dynamic or partially unknown environments, adding explicit robot collision avoidance, and evaluating performance with more robots or more complex maps. Running a larger number of tests on different randomly generated scenarios would also help confirm the consistency and reliability of the algorithms in real search and rescue settings.

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